

[illegible]

Theory				Goals										Tools														Surveys		Total			
Stateless		Stateful		Attacker Analysis				Trends		Attack Analysis		Deception Strategy		Actively Maintained		Current Status		Source-code availability				Purpose		AI Integration									
				Basic				Advanced		None		Protect production		None		Not Maintained		Open-source		Not Open-source		Research		Production		None		ML					
Source	Count	Source	Count	Source	Count	Source	Count	Source	Count	Source	Count	Source	Count	Source	Count	Source	Count	Source	Count	Source	Count	Source	Count	Source	Count	Source	Count	Source	Count	Source	Count		
AL	59	AL	19	AL	40	AL	12	AL	20	AL	55	AL	23	AL	8	AL	10	AL	18	AL	27	AL	37	AL	8	AL	30	AL	11	AL	2		
OL	2	OL	2	OL	17	OL	7	OL	5	OL	13	OL	19	OL	5	OL	6	OL	7	OL	8	OL	11	OL	8	OL	11	OL	3	OL	7		
Year (SL)		Year (SL)		Year (SL)		Year (SL)		Year (SL)		Year (SL)		Year (SL)		Year (SL)		Year (SL)		Year (SL)		Year (SL)		Year (SL)		Year (SL)		Year (SL)		Year (SL)		Year (SL)			
2003	2	2017	1	2003	1	2015	1	2009	1	2009	2	2003	2	2009	2	2003	2	2009	2	2003	2	2009	2	2009	2	2003	2	2009	2	2015	1	2016	1
2006	2	2006	1	2007	1	2017	1	2015	2	2020	1	2015	2	2015	2	2022	2	2015	2	2015	2	2022	2	2015	2	2024	1	2015	1	2017	2	2011	1
2015	2	Year (AL)		2017	1	2020	1	2017	2	2022	3	2017	2	2017	2	2024	2	2017	2	2017	2	2025	3	2017	1	2024	2	2025	1	2018	1	2012	1
2017	1	2011	1	2019	1	2021	2	2025	2	2019	1	2023	2	2019	1	2025	3	2019	1	2019	1	2025	3	2019	1	2024	1	2019	1	2020	1	2014	1
2019	1	2012	1	2022	2	2022	1	Year (AL)		2020	1	2024	3	2020	1	2024	3	2020	1	2020	1	2022	1	2020	1	2020	1	2020	2	2022	2	2011	1
2020	2	2014	1	2023	3	2024	1	2021	2	2021	2	2025	8	2009	1	2009	1	2016	1	2016	1	2022	2	2018	1	2024	1	2022	1	2022	1	2016	4
2021	2	2016	1	2024	4	Year (AL)		2012	1	2022	1	Year (AL)		2015	1	2015	1	2017	5	2011	1	2016	1	2024	1	2019	1	2025	2	2020	2	2017	11
2022	4	2018	1	2025	4	2016	2	2014	1	2023	1	2017	2	2022	2	2017	2	2018	1	2014	1	2017	5	Year (AL)		2020	1	2018	1	2024	1	2018	5
2023	3	2019	2	Year (AL)		2017	3	2016	1	2024	1	2018	2	2024	2	2019	1	2019	3	2016	1	2018	1	2024	1	2024	1	2019	3	2025	3	2019	15
2024	4	2020	4	2024	1	2019	2	2018	1	Year (AL)		2019	4	2025	3	2020	1	2020	6	2018	3	2019	3	2011	1	2025	2	2016	2	2020	1	2020	17
2025	7	2021	1	2011	1	2020	2	2019	4	2004	1	2020	4	2004	1	2020	4	2004	1	2020	4	2020	6	2012	1	2017	5	2021	1	2021	1	2021	8
Year (AL)		Count		2022	3	2021	1	2020	4	2006	1	2021	2	2016	1	2004	1	2023	3	2020	1	2022	3	2014	1	2018	3	2022	2	2018	3	2022	11
2004	1	2023	2	2014	1	2022	1	2021	1	2011	1	2022	1	2019	2	2011	1	2024	2	2021	2	2023	3	2016	2	2017	3	2019	4	2023	9	2023	9
2006	1	2024	1	2017	2	2025	1	2022	2	2012	1	2024	5	2020	1	2014	1	2025	2	2022	2	2024	2	2017	3	2020	6	2020	6	2024	12	2024	12
2016	2	2025	1	2018	4	2003	0	2023	1	2014	1	2025	3	2021	1	2018	3	2024	1	2018	3	2016	3	2024	1	2019	6	2022	5	2021	1	2025	15
2017	7	2003	0	2019	6	2004	0	2024	3	2016	3	2018	3	2022	1	2019	2	2025	1	2019	2	2024	1	2021	1	2019	6	2022	5	2024	1	2025	15
2018	6	2004	0	2020	5	2005	#N/A	2003	0	2017	5	2018	5	2020	1	2016	1	2004	1	2023	3	2019	4	2020	6	2020	6	2020	6	2024	1	2025	15
2019	12	2005	#N/A	2021	2	2006	#N/A	2004	0	2018	5	2020	5	2021	1	2018	3	2024	1	2018	3	2016	3	2024	1	2019	6	2022	5	2024	1	2025	15
2020	11	2006	#N/A	2022	5	2009	0	2005	#N/A	2019	10	2020	10	2022	1	2019	2	2025	1	2019	2	2024	1	2021	2	2020	6	2020	6	2024	1	2025	15
2021	4	2009	0	2023	3	2011	#N/A	2006	#N/A	2020	11	2020	11	2022	2	2020	2	2024	2	2022	2	2024	2	2017	3	2020	6	2020	6	2024	1	2025	15
2022	4	2011	#N/A	2024	5	2012	#N/A	2009	50	2021	3	2021	3	2022	2	2020	2	2024	2	2022	2	2024	2	2017	3	2020	6	2020	6	2024	1	2025	15
2023	1	2012	#N/A	2025	4	2014	#N/A	2011	#N/A	2022	6	2022	6	2023	1	2018	3	2024	1	2023	3	2019	3	2024	1	2020	6	2020	6	2024	1	2025	15
2024	6	2014	#N/A	2026	4	2015	#N/A	2012	#N/A	2023	3	2023	3	2024	1	2019	2	2025	1	2024	1	2020	6	2020	6	2020	6	2020	6	2024	1	2025	15
2025	4	2015	#N/A	2027	4	2016	#N/A	2013	#N/A	2024	3	2024	3	2025	1	2020	4	2025	1	2025	1	2020	6	2020	6	2020	6	2020	6	2024	1	2025	15
2003		100		2016	#N/A	2005	#N/A	2017	50	2015	50	2015	50	Year (AL)		2016	1	2004	1	2023	3	2019	4	2020	6	2020	6	2020	6	2024	1	2025	15
2004		0		2017	50	2006	#N/A	2018	50	2016	50	2016	50	Year (AL)		2017	1	2005	1	2024	4	2020	6	2020	6	2020	6	2020	6	2024	1	2025	15
2005		#N/A		2018	#N/A	2009	50	2019	50	2017	0	2017	0	Year (AL)		2018	1	2006	1	2025	5	2020	6	2020	6	2020	6	2020	6	2024	1	2025	15
2006		#N/A		2019	0	2011	#N/A	2020	50	2018	50	2018	50	Year (AL)		2019	1	2007	1	2026	6	2020	6	2020	6	2020	6	2020	6	2024	1	2025	15
2009		100		2020	0	2012	#N/A	2021	100	2019	0	2019	0	Year (AL)		2020	1	2008	1	2027	7	2020	6	2020	6	2020	6	2020	6	2024	1	2025	15
2011		#N/A		2021	0	2014	#N/A	2022	25	2020	0	2020	0	Year (AL)		2021	1	2009	1	2028	8	2020	6	2020	6	2020	6	2020	6	2024	1	2025	15
2012		#N/A		2022	0	2015	0	2023	0	2021	0	2021	0	Year (AL)		2022	1	2010	1	2029	9	2020	6	2020	6	2020	6	2020	6	2024	1	2025	15
2014		#N/A		2023	0	2016	#N/A	2024	0	2022	0	2022	0	Year (AL)		2023	1	2011	1	2030	10	2020	6	2020	6	2020	6	2020	6	2024	1	2025	15
2015		100		2024	0	2017	50	2025	12.5	2023	33.33	2023	33.33	Year (AL)		2024	1	2012	1	2031	11	2020	6	2020	6	2020	6	2020	6	2024	1	2025	15
2016		#N/A		2025	12.5	2018	#N/A	2026	0	2024	0	2024	0	Year (AL)		2025	1	2013	1	2032	12	2020	6	2020	6	2020	6	2020	6	2024	1	2025	15
2017		50		2026	0	2019	100	2027	25	2025	25	2025	25	Year (AL)		2026	1	2014	1	2033	13	2020	6	2020	6	2020	6	2020	6	2024	1	2025	15
2018		#N/A		2027	0	2020	0	2028	0	2026	0	2026	0	Year (AL)		2027	1	2015	1	2034	14	2020	6	2020	6	2020	6	2020	6	2024	1	2025	15
2019		100		2028	0	2021	0	2029	0	2027	0	2027	0	Year (AL)		2028	1	2016	1	2035	15	2020	6	2020	6	2020	6	2020	6	2024	1	2025	15
2020		100		2029	0	2022	50	2030	#N/A	2028	0	2028	0	Year (AL)		2029	1	2017	1	2036	16	2020	6	2020	6	2020	6	2020	6	2024	1	2025	15
2021		100		2030	0	2023	100	2031	0	2029	0																						
2022		100		2031	100	2024	100	2032	0	2029	0																						
2023		100		2032	100	2025	50	2033	0	2031	0																						
2024		100		2033	100	2026	100	2034	0	2032	0																						
2025		87.5		2034	100	2027	100	2035	100	2033	100																						
2026		87.5		2035	100	2028	100	2036	100	2034	100																						
2027		87.5		2036	100	2029	100	2037	100	2035	100																						
2028		87.5		2037	100	2030	100	2038	100	2036	100																						
2029		87.5		2038	100	2031	100	2039	100	2037	100																						
2030		87.5		2039	100	2032	100	2040	100	2038	100																						
2031		87.5		2040	100	2033	100	2041	100	2039	100																						
2032		87.5		2041	100	2034	100	2042	100	2040	100																						
2033		87.5		2042	100	2035	100	2043	100	2041	100																						
2034		87.5		2043	100	2036	100	2044	100	2042	100																						
2035		87.5		2044	100																												

			Tags																													
Study	Year	Origin	AI	ML	LLM	RL	IRL	Attack Analysis	Botnet	DoS	Ransomware	Blockchain	Conversion	Deployment	Tool	Framework	Energy	Fingerprinting	Network	Darknet	Reputation	GAN	MTD	CMD	Zero-Trust	SDN	TSP	Game-Theory	MDP	SoLA	Smart Cities	Survey
Shafiq, M., Valent	2024	AL	x	x	x										x																	
Ozal, H. T., & Car	2024	AL	x	x	x										x																	
Tencent Cloud (,	2024	GL																														
Trapster (2024),	2024	GL																														
GCA (2024), Ely	2024	GL													x																	
Acalvio (2024), I	2024	GL													x																	
Mont, Z., David,	2025	AL						x											x													x
Guan, C., & Cao,	2025	AL												x	x											x						
Lanz, S., Pignol,	2025	AL	x	x			x																									x
Valentini, E. P., F	2025	AL						x																								
Oroum, Y., Asad,	2025	AL	x	x							x			x		x													x			
Kouari, O. el, Las	2025	AL												x																		
Comney, D., Nik	2025	AL																														
SecurityHive (20	2025	GL	x	x										x		x																
Arzen (2026), M	2025	GL																														
Netlas (2026), B	2025	GL																														
Palo Alto (2026)	2025	GL																														x
Fidels Security (	2025	GL												x	x																	
Zacaler (2026) (	2025	GL																														
Tractile Tech (20	2025	GL	x																							x						
Hoguero, L. O. (	2025	GL	x	x	x		x															x										
Count			24	22	4	10	1	25	9	4	1	2	1	18	58	3	2	3	2	1	1	1	6	1	1	8	1	11	4	1	2	14